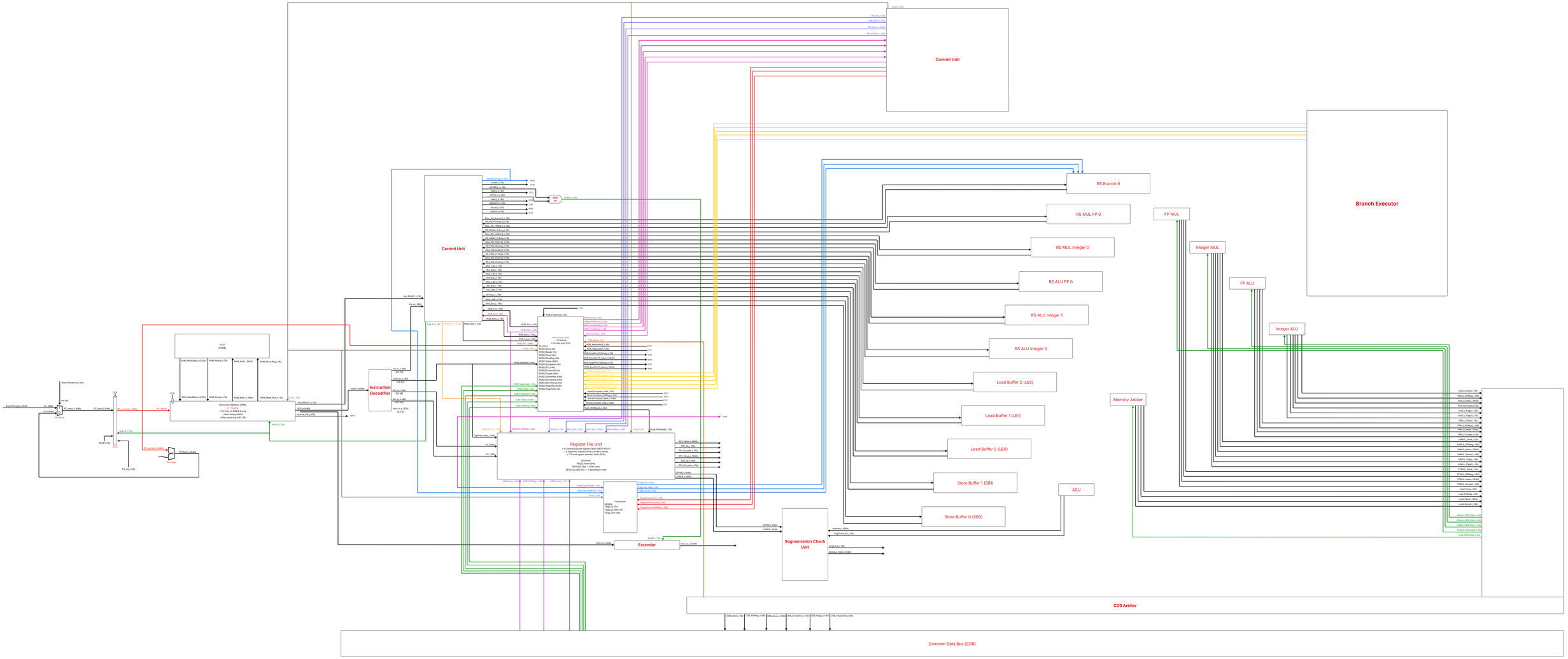
Commit Unit
Branch Executor
RS Branch 0
RS MUL FP 0
RS MUL Integer 0
RS ALU FP 0
RS ALU Integer 1
RS ALU Integer 0
Load Buffer 2 (LB2)
Load Buffer 1 (LB1)
Load Buffer 0 (LB0)
Store Buffer 1 (SB1)
Store Buffer 0 (SB0)
FP MUL
Integer MUL
FP ALU
Integer ALU
Memory Arbiter
AGU
Control Unit
Instruction Decodifier
Register File Unit
Segmentation Check Unit
Extender
CDB Arbiter
Common Data Bus (CDB)
RAM
Instruction Memory SRAM
Reorder Buffer (ROB)
Flags Register
PC Mux
PC Adder
RESET (1b)
PC_next (32b)
PC_current (32b)
BranchTarget (32b)
Instruction
Decodifier
Register File Unit
Segmentation
Check Unit
Extender
Flags Register
Commit Unit
Branch
Executor
Memory
Arbiter
Integer
ALU
Integer
MUL
FP
ALU
FP
MUL
RS
Branch
0
RS
ALU
Integer
0
RS
ALU
FP
0
RS
ALU
Integer
1
RS
MUL
Integer
0
RS
MUL
FP
0
Load
Buffer
2
(LB2)
Load
Buffer
1
(LB1)
Load
Buffer
0
(LB0)
Store
Buffer
1
(SB1)
Store
Buffer
0
(SB0)
AGU
CDB
Arbiter
Common
Data
Bus
(CDB)